

CENG381 Stochastic Processes

Random Mapping Representation — Practice 1 (Answer Key)

Q1 Solution

a) Transition matrix P for C_3 :

	0	1	2
0	[0 1/2 1/2]		
1	[1/2 0 1/2]		
2	[1/2 1/2 0]		

b) Verification of the random mapping $f(j,Z) = (j+Z) \bmod 3$:

For each starting state j , check all reachable states k :

$$\begin{aligned} j=0: & \quad P\{f(0,Z)=1\} = P\{(\theta+Z) \bmod 3 = 1\} = P\{Z=1\} = 1/2 = P(0,1) \quad \checkmark \\ & \quad P\{f(0,Z)=2\} = P\{(\theta+Z) \bmod 3 = 2\} = P\{Z=-1\} = 1/2 = P(0,2) \quad \checkmark \\ & \quad P\{f(0,Z)=0\} = P\{(\theta+Z) \bmod 3 = 0\} = 0 \text{ (Z is never } \theta) \quad \checkmark \\ j=1: & \quad P\{f(1,Z)=2\} = P\{Z=+1\} = 1/2 = P(1,2) \quad \checkmark \\ & \quad P\{f(1,Z)=0\} = P\{Z=-1\} = 1/2 = P(1,0) \quad \checkmark \\ j=2: & \quad P\{f(2,Z)=0\} = P\{Z=+1\} = 1/2 = P(2,0) \quad \checkmark \\ & \quad P\{f(2,Z)=1\} = P\{Z=-1\} = 1/2 = P(2,1) \quad \checkmark \end{aligned}$$

Valid representation. $\checkmark$

c) Computing $P(X_1=1)$ and $P(X_2=0)$:

$$\begin{aligned} P(X_1=1 \mid X_0=0) &= P\{(\theta+Z_1) \bmod 3 = 1\} = P\{Z_1=1\} = 1/2 \\ P(X_2=0 \mid X_0=0) &= P(0,0) \cdot P^{11}_{\text{entry}} \dots \text{ use } P^2: \\ p_{\{00\}}^{(2)} &= P(0,1) \cdot P(1,0) + P(0,2) \cdot P(2,0) \\ &= (1/2)(1/2) + (1/2)(1/2) = 1/4 + 1/4 = 1/2 \\ \text{So } P(X_2=0) &= 1/2. \end{aligned}$$

Q2 Solution

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a) Random mapping representation for C_4 :

Define $f(j, Z) = (j + Z) \bmod 4$ with Z uniform on $\Lambda = \{-1, +1\}$.

Verification (same logic as Q1 for C_3):

$$\begin{aligned} P\{f(j, Z) = j+1 \bmod 4\} &= P\{Z=+1\} = 1/2 = P(j, j+1 \bmod 4) \quad \checkmark \\ P\{f(j, Z) = j-1 \bmod 4\} &= P\{Z=-1\} = 1/2 = P(j, j-1 \bmod 4) \quad \checkmark \end{aligned}$$

b) Closed-form expression for X_n :

$$\begin{aligned} X_1 &= (X_0 + Z_1) \bmod 4 = (0 + Z_1) \bmod 4 = Z_1 \bmod 4 \\ X_2 &= (X_1 + Z_2) \bmod 4 = (Z_1 + Z_2) \bmod 4 \\ &\dots \\ X_n &= (Z_1 + Z_2 + \dots + Z_n) \bmod 4 = S_n \bmod 4 \end{aligned}$$

c) Coupling argument and convergence:

With $X_0=0$ and $Y_0=2$ driven by the same noise sequence $Z_1, Z_2, \dots$:

$$\begin{aligned} X_n &= S_n \bmod 4 \\ Y_n &= (2 + S_n) \bmod 4 \\ \\ X_n - Y_n &= S_n - (2 + S_n) = -2 \pmod{4} \quad \text{for ALL } n \geq 0. \end{aligned}$$

The two chains never couple: they maintain a constant offset of 2 modulo 4 for every realisation of the noise.

Implication: because the C_4 random walk has period 2, using the same noise does not bring together paths started 2 apart. A standard coupling proof of convergence requires coupling, but here the chains never meet. This is consistent with the fact that the C_4 walk is PERIODIC (period 2) and does NOT converge to π . To fix this, one would use the lazy chain or a different coupling.

Q3 Solution

a) Continuous random mapping representation:

For each state x , partition $[0,1]$ into intervals proportional to transition probabilities:

$$\begin{aligned} f(0, z): \quad & z \text{ in } [0, 1/3) \rightarrow \text{next state } 0 \\ & z \text{ in } [1/3, 2/3) \rightarrow \text{next state } 1 \\ & z \text{ in } [2/3, 1] \rightarrow \text{next state } 2 \end{aligned}$$

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f(1, z):  z in [0, 1/2) → next state 0
           z in [1/2, 1] → next state 1

f(2, z):  z in [0, 1/2) → next state 1
           z in [1/2, 1] → next state 2
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Since $Z \sim \text{Uniform}[0,1]$, $P\{f(x,Z)=y\} = \text{length of interval} = P(x,y)$. ✓

(This is the standard 'quantile / inverse-CDF' representation.)

b) Irreducibility, aperiodicity, stationary distribution:

Irreducible: $0 \rightarrow 1$ (direct), $0 \rightarrow 2$ (direct), $1 \rightarrow 0$ (direct), $2 \rightarrow 1 \rightarrow 0$ (2 steps). 2 can reach 0 in 2 steps. All states communicate. ✓

Aperiodic: $P(0,0)=1/3 > 0$ and $P(1,1)=1/2 > 0$, so both 0 and 1 have self-loops $\rightarrow T(0)$ contains $t=1 \rightarrow \text{period}=1$. ✓

Balance equations $\pi \cdot P = \pi$:

$$\pi_0 = (1/3)\pi_0 + (1/2)\pi_1 \quad \dots \text{ (I)}$$

$$\pi_1 = (1/3)\pi_0 + (1/2)\pi_1 + (1/2)\pi_2 \quad \dots \text{ (II)}$$

$$\pi_2 = (1/3)\pi_0 \quad + (1/2)\pi_2 \quad \dots \text{ (III)}$$

$$\pi_0 + \pi_1 + \pi_2 = 1$$

From (I): $(2/3)\pi_0 = (1/2)\pi_1 \rightarrow \pi_1 = (4/3)\pi_0$

From (III): $(1/2)\pi_2 = (1/3)\pi_0 \rightarrow \pi_2 = (2/3)\pi_0$

Sum: $\pi_0 + (4/3)\pi_0 + (2/3)\pi_0 = \pi_0 \cdot (3/3 + 4/3 + 2/3) = 3\pi_0 = 1$

$$\pi = (1/3, \quad 4/9, \quad 2/9) \approx (0.333, \quad 0.444, \quad 0.222)$$

c) Two-step distribution from $X_0=0$:

$$\mu_0 = (1, 0, 0) \quad (\text{start at state } 0)$$

$$\mu_1 = \mu_0 \cdot P = (1/3, \quad 1/3, \quad 1/3)$$

$$\mu_2[0] = (1/3)(1/3) + (1/3)(1/2) + (1/3)(0) = 1/9 + 1/6 = 2/18 + 3/18 = 5/18$$

$$\begin{aligned} \mu_2[1] &= (1/3)(1/3) + (1/3)(1/2) + (1/3)(1/2) = 1/9 + 1/6 + 1/6 = \\ &2/18 + 3/18 + 3/18 = 8/18 \end{aligned}$$

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$$\mu_2[2] = (1/3)(1/3) + (1/3)(0) + (1/3)(1/2) = 1/9 + 0 + 1/6 = 2/18 + 3/18 = 5/18$$

$$\mu_2 = (5/18, 8/18, 5/18) \approx (0.278, 0.444, 0.278)$$

Compare to $\pi = (0.333, 0.444, 0.222)$. After only 2 steps, $\mu_2[1]$ already equals $\pi_1 = 4/9$ exactly, while $\mu_2[0] = 5/18 \approx 0.278$ vs $\pi_0 = 1/3 \approx 0.333$. The chain is converging but not yet fully mixed after 2 steps.